

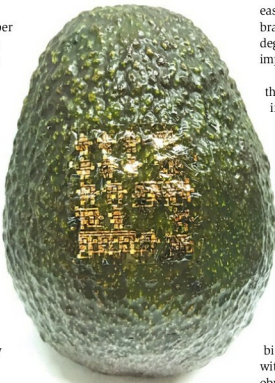
wireless communication could deliver an average power of $0.23\mu\text{W}$ per square-mm of electrode area for an average of 6.1 days, when used for temperature measurements in the gastrointestinal tract of pigs.

In yet another development, Dr Christopher Bettinger of Carnegie-Mellon University proposed biodegradable elastomers as structural polymers, and melanin-based pigments as materials for on-board energy storage in implantable and ingestible electronics. His lab focuses on the development of biomaterials-based micro-electromechanical systems (MEMS) for use in regenerative medicine, neural interfaces, drug delivery, etc. They have developed edible, biocompatible batteries that use non-toxic materials already present in the body, with available liquids such as stomach acid as the electrolyte. Their cathodes use melanin, while anodes are made of manganese oxide. These electrodes dissolve harmlessly after use.

A team of researchers, including Zhaowei Guo and others from Fudan University in China, have developed a family of flexible and biocompatible aqueous sodium-ion batteries for implants. Instead of toxic electrolytes, these batteries use sodium-containing aqueous electrolytes such as normal saline and cell-culture medium. The cell-culture medium comprises amino acids, sugars and vitamins, which is quite similar to the fluid that surrounds cells in the human body.

The team made two kinds of batteries. One was a two-dimensional belt-shaped battery made of thin films of electrode material stuck on a mesh made of steel strands. The other used a woven carbon nanotube fibre backbone with embedded nanoparticle electrode materials.

This fibre-shaped electrode with normal saline or cell-culture medium electrolyte surprised the researchers by accelerating the conversion of dissolved oxygen into hydroxide



A new flexible and biodegradable semiconductor developed at Stanford University that can adhere to smooth or rough surfaces. Here it is shown clinging to an avocado! (Courtesy: Bao Lab)

ions, and changing the pH. While this could be detrimental to the effectiveness of the battery, it could be very useful in biological and medical investigations like cancer starvation therapy.

In Stanford, a team led by Zhenan Bao has developed a flexible electronic device that can easily be degraded into non-toxic compounds just by adding a weak acid like vinegar to it. Apart from the polymer and the degradable electronic circuit, the team has also developed a new cellulose-based biodegradable substrate material for mounting the electrical components. According to them, this substrate supports electrical components, flexing and moulding to rough and smooth surfaces alike.

Bao is not new to this field. Human skin has always fascinated her, and she previously developed a skin-inspired stretchable electrode, which was so flexible that it could

easily interface with the skin or brain. However, the electrode's non-degradability made it unsuitable for implantable devices.

Bao says in a media report that they came up with an idea of making these molecules using a special type of chemical linkage that can retain the ability for the electron to smoothly transport along the molecule. But this chemical bond is sensitive to weak acid—even weaker than pure vinegar. So the result was a polymer material that could not only carry an electronic signal but also break down easily to product concentrations much lower than the published acceptable levels found in drinking water.

How do sensors made of biocompatible materials compare with those made of food itself? Well, obviously you would prefer the latter. So thought a team of researchers from the University of Wollongong in Australia! In a paper, they reveal that the development of highly swollen, strong, conductive hydrogel materials is necessary for the advancement of edible devices. So they analysed the electrical properties of everyday food products like jelly, Vegemite and Marmite (two popular brands of yeast extract) and harnessed them to make edible hydrogel electrodes. These gels were used to demonstrate a capacitive pressure sensor that can help detect digestive pressure abnormalities such as intestinal motility disorders.

Eat your robot

We have biocompatible materials as well as transistors, sensors, batteries, electrodes and capacitors made using such materials. So you might think, why not a robot, too? But, what sets a robot apart from other computing systems is its ability to move—a robot needs an actuator, and attempts at making an edible one have been unpalatable all along!

Switzerland-based research